S-R-44D
August 11, 1970
SUPERSEDING
Fed. Spec. S-R-44c
July 28, 1967

FEDERAL SPECIFICATION

RANGES, GAS (DOMESTIC)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers cabinet type, flush to wall, free-standing and slide-in domestic gas ranges with a top cooling section, oven(s), and a broiler.

1.2 Classification. Gas ranges shall be of the following types, sizes, and styles as specified (see 6.2).

Type I	-	Single oven.
Size 20	-	20 inch range width.
Size 30	-	30 inch range width.
Size 36	-	36 inch range width.
Style 1	-	Free standing.
Style 2	-	Slide-in.
Type II	-	Double oven (free standing).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

FSC 7310

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-R-11337 - Ranges, Packaging and Packing of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of U. S. Military Property.

(Copies of Military Specifiations and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American National Standards Institute, Inc. (ANSI) Standard:

Z 21.1.1 - Approval Requirements for Domestic Gas Ranges

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018, or the American Gas Association, Inc., 8501 East Pleasant Valley Road, Cleveland, Ohio 44131.)

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American Gas Association, Inc. (AGA) Publication:

* Directory of Certified Appliances and Accessories (1968).

(Application for copies should be addressed to the American Gas Association, Inc., 8501 East Pleasant Valley Road, Cleveland, Ohio 44131.)

American Society for Testing and Materials (ASTM) Standards:

C 282 - Tentative Method of Test for Acid Resistance of Porcelain Enamels.

C 286 - Terms Relating to Porcelain Enamel.

D 635 - Flammability of Rigid Plastics Over 0.127 CM (0.050 Inch) in Thickness.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

3. REQUIREMENTS

3.1 First article. This specification contains provisions for first article test and approval (see 4.2, 6.2 and 6.4).

3.2 Standard product. The ranges delivered under this specification shall be the manufacturer's standard commercial product, except for any changes necessary to comply with specification requirements. All ranges of one type, size, and style furnished under this specification for any one contract, including parts and assemblies thereof, shall be interchangeable. No deviations shall be acceptable without prior approval of the procuring agency or authorized representative thereof.

3.3 Safety standard. The gas ranges shall conform to ANSI Z 21.1.1.

3.3.1 Certification. Prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the ranges conform to the requirements of ANSI Z 21.1.1. Acceptable evidence of meeting the applicable requirements of ANSI Z 21.1.1 shall be a photostatic copy of the AGA Appliance Certificate; a listing of the range in the AGA "Directory of Certified Appliances and Accessories"; or a certified test report from a recognized, independent testing laboratory acceptable to the Government stating that the ranges have been tested and conform to ANSI Z 21.1.1.

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3.4 Design. Ranges shall be designed for flush to wall installation and for operation with natural gas and liquefied petroleum gas. Style 1 ranges shall be free standing and style 2 ranges shall be slide-in. Type I, styles 1 and 2 ranges shall be nominal 20, 30, or 36 inches wide as stated in 1.2 and shall be provided with one oven. Type II ranges shall be free standing and shall be nominal 40 inches wide with two ovens. Ovens shall be capable of maintaining a temperature as low as 250°F. (see 4.4.1).

3.4.1 Dimensions. Ranges shall conform to the dimensional requirements specified in table I.

TABLE I. Dimensions

Measurements	Type I (styles 1 and 2)			Type II
	Size 20	Size 30	Size 36	
	Inches	Inches	Inches	Inches
Width from end to end 1/				
Style 1	19 to 22	29 to 32	36 to 38	39 to 41
Style 2	20	30	36	
Depth from front to back, excluding handles	22 to 28	22 to 28	22 to 28	24 to 29
Height 2/ from floor to cooking top (plus or minus 1/4 inch)	36	36	36	36
Baking oven interior				
Width 3/ (min.)	16	16	16	16 each
Depth (min.)	16-1/2	16-1/2	16-1/2	18
Height (min.)	12-1/2	12-1/2	12-1/2	13

1/ Width tolerance plus or minus 1/4 inch for style 2.

2/ On depressed tops, measurement is to top of raised edges.

3/ Sidewall to sidewall dimension.

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3.4.2 Special characteristics.

3.4.2.1 Type I (single oven) ranges. When specified (see 6.2), type I ranges shall be provided with additional features specified in 3.4.2.2 except for the two broiler compartments. Timers and clocks when furnished shall be tested as specified in 4.4.1 to determine proper operation.

3.4.2.2 Type II (double oven) range. The type II range shall be provided with the following features in addition to the requirements specified herein:

(a) Appliance outlet.

(b) Interval timer.

(c) Automatic electric clock control for at least one oven (a combination oven timing clock and interval timer is acceptable).

(d) Two broiler compartments, one of which may be a meat oven.

(e) Back splash light.

Timers and clocks shall be tested as specified in 4.4.1 to determine proper operation.

3.5 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for his ranges provided the completed items comply with all provisions of this specification. Commercial tolerances are applicable to all metal thicknesses.

3.6 Construction.

3.6.1 Broiler. The broiler(s) shall be a smokeless type (no smoke issue from flue) and shall be located below the oven. Broilers shall be provided with a broiler pan and a removable grill or a smokeless grill pan. The broiler shall be a type that slides out of the compartment when a door is opened or a drawer is withdrawn or of a type that remains stationary when the door is open. Doors for movable broilers shall be drop-type and pull-out drawers shall be mounted on nylon wheels without bearings, or ball or roller bearing glides, slides, or wheels. Doors for stationary broilers shall be either a drop-type or side-opening type. Broiler vent discharge openings shall be located to deflect flue gases away from rear and side walls.

3.6.2 Oven(s). Oven(s) shall conform to the dimensions specified in table I. The oven vent shall deflect smoke and flue gases away from rear

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and side walls. Oven bottoms shall be removable for cleaning without the use of tools. The oven shall be equipped with a thermostat which will provide automatic temperature control of the oven. The oven thermostat shall be provided with a graduated scale to a high of not less than 500°F. The oven thermostat control shall have the bake and broil positions of temperature legibly and permanently marked. Ovens shall be provided with a light. Where oven doors contain glass vision panels, oven lights shall be switch controlled.

3.6.2.1 Oven racks. The baking oven racks shall be constructed of corrosion-resisting steel, porcelain enameled steel, or chrome or nickel plated steel. Oven racks shall be provided with stops which will permit the rack to be withdrawn at least one-half its depth. When tested as specified in 4.4.4, the rack shall not deflect more than 1/8 inch for racks extending up to 8 inches and the rack shall not deflect more than 1/4 inch for racks extending up to 11 inches.

3.6.3 Oven and broiler doors. Oven and broiler doors shall be provided with springs tensioned to hold the doors closed. Drop shelf type doors except doors on broilers located under the oven, shall be counterbalanced and when fully opened shall remain in that position without securing. Means shall be provided to adjust the oven door to maintain the door against the oven seal. When tested as specified in 4.4.2, the door shall fit closely to the oven facing. The oven door for the style 2 range shall be recessed and the sides of the range cabinet shall project past the oven door. The sides of the style 1 and type II range cabinet shall be flush with or shall not extend more than 1/8 inch beyond the outside surface of the door in the closed position. Side trim strips may be installed on the range cabinet sides so that the trim strips project past the oven door.

3.6.4 Fuel system. Components of the fuel system shall be assembled so there are no leaks when tested as specified in 4.4.7. Burner valve control knobs and handles shall not bind or be loose when tested as specified in 4.4.1. Gas valves for ovens, top surface burners, and pilot gas valves on ranges shall be provided with universal orifices or two interchangeable orifice hoods or bushings. When hoods or bushings are furnished each hood or bushing shall be drilled for one of the two types of gas specified in 3.4 and shall be identified in accordance with the manufacturer's standard practice. Instructions for conversion from one gas to the other shall be furnished. The pilot lines to the top burners shall be routed to allow for ease in cleaning of burner tray or box bottom.

3.6.5 Top burners. Types I and II ranges shall be provided with four top burners. At least one of the top burners on type I ranges and at least two of the top burners on type II ranges shall be rated at minimum 12,000 British

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Thermal Units (BTU) per hour for natural gas and minimum 9000 BTU per hour for liquefied petroleum gas. Burner arrangement shall be in accordance with the manufacturer's standard practice. Except for Navy procurements (see 6.2) ranges shall have at least one of the top burners controlled by a thermostatic device. Burner heads shall fit flush with the body when heads are two or three piece construction and shall be finished as specified in 3.8.5. Burner and caps shall be aluminum, cast iron, or steel finished as specified in 3.9.5 except when procurements for the Far East or Pacific locations are specified (see 6.2), burners shall be cast iron or corrosion-resisting steel.

3.6.6 Grates. Each top burner shall be provided with a cast iron or steel grate. The center of the grate shall not be higher than the outside supporting periphery of the grate, and the vertical displacement at any point on any grate shall not be more than 1/8 inch when tested as specified in 4.4.3. Grates shall be finished as specified in 3.9.2.

3.6.7 Aeration bowls and burner trays. When aeration bowls are provided, they shall be removable to permit cleaning or shall be incorporated as part of a removable or hinged cooking top. Aeration bowls shall be corrosion-resisting steel, chrome or nickel plated steel, or steel coated with procelain enamel. Removable aeration bowls and burner trays when furnished shall be finished as specified in 3.9.3 and 3.9.4, respectively.

3.6.8 Knobs and handles. Plastic material for burner valve control knobs and door and drawer shall be molded and heat-resistant. The various positions shall be legibly and permanently marked on or immediately adjacent to each valve control knob. Controls or knobs shall not be installed on the cooking surface between burners. When tested as specified in 4.4.5.1 and 4.4.5.2 the plastic knobs shall be self-extinguishing and shall not break, chip, or crack.

3.6.9 Back splash. Each range shall be equipped with a back splash not less than 4 inches higher than the top cooking surface and extending the full width of the range top. When a glass back splash is furnished, the glass shall be heat tempered.

3.6.10 Insulation. Blanket-type, mineral or fiberglass insulation shall be used throughout the range. Insulation shall be installed to prevent settling.

3.6.11 Leveling devices. Each range shall be provided with at least two leg leveling devices.

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3.6.12 Electrical equipment. Unless otherwise specified (see 6.2), timers, lights, clocks and other electrical equipment on the ranges shall operate on a nominal 120 volt, 60 hertz (H_2) 1-phase system.

3.7 Accessories. Accessories normally furnished as part of the manufacturer's standard product but in excess of those specified herein, will be acceptable. Removal of accessories that affect appearance or serviceability is not acceptable.

3.8 Operation. When tested as specified in 4.4.1, burners and pilots shall operate as specified in ANSI Z 21.1.1.

3.9 Finish.

3.9.1 Exterior surfaces. The exterior surfaces of cabinet tops, oven doors, broiler door or drawer, compartment door panels, and front panels, shall have a white porcelain enamel finish. Side panels shall have a white porcelain enamel finish or a white, high temperature baked enamel finish. The back splash, back panel, baffle plate, trim, (and toe-space panel, when provided) shall be finished in accordance with the manufacturer's standard practice. The porcelain enamel on the cooking top shall have a class A or better acid resistance when tested in accordance with ASTM Method C 282 (see 4.3.2), **and shall not be thicker than 12 mils when tested as specified in 4.4.6. The procelain enamel shall be glossy and free from discolorations, cracks, thin spots, holes, burn-off, chipping, copperheads, fish scaling, or enamel lumps, i.e., wrinkles or wavy or other imperfections.**

3.9.2 Grates. Surfaces of the cast iron and carbon steel grates which are visible when in normal operating position shall be coated with porcelain enamel. The finished enamel surfaces shall be free from pinholes, blisters, and lumps of enamel.

3.9.3 Aeration bowls. When porcelain enamel finish is provided on aeration bowls it shall have a maximum thickness of 12 mils when tested as specified in 4.4.6.

3.9.4 Oven linings, broiler linings, and burner trays. The oven linings, broiler linings, and burner trays shall have a porcelain enamel finish and shall have a maximum thickness of 8 mils when tested as specified in 4.4.6.

3.9.5 Burners. Burners and caps fabricated of aluminum shall have a natural metal or anodized finish. Cast iron and steel burners and caps shall be porcelain enameled, chrome or nickel plated, or have a baked enamel finish. Burner box shall have a porcelain enameled, baked enamel or aluminized steel finish.

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3.9.6 Unexposed surfaces. Surfaces of parts not exposed to view in normal usage shall be protected in such a manner that any oxidation which may occur will not interfere with its intended function. An oxide coating is acceptable for these unexposed surfaces.

3.10 Marking for identification. The ranges shall be permanently marked with the serial and model number and with the manufacturer's name, trade name, or trademark of such known characters as to readily identify the manufacturer. In addition, any information as required by ANSI Z 21.1.1 shall also appear on the nameplate.

3.11 Workmanship. The finished ranges shall be clean and shall not contain any burrs, slivers, sharp edges or rough die, tool, gouge or grind marks. Holes for bolts, screws, rivets, and threaded fasteners shall be accurately punched or drilled and any obstruction that prevents proper seating of rivets shall be removed. Driven rivets shall be fully peened with the heads in full contact with riveted members. Components shall not be fractured, buckled, kinked, bent, dented, punctured, or malformed. There shall be no evidence of pits or stains, blistering, flaking, peeling, or large discoloration (exposed surfaces). Components shall be free of sags, runs, areas of no film, rust, imbedded foreign matter, and wet or tacky painted surfaces.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First article inspection. When a preproduction sample is required it shall be examined for the defects listed in table II and dimensions specified and tested in accordance with 4.4.1 through 4.4.4 and 4.4.7. Failure to pass any of the tests shall be cause for rejection of the preproduction sample. The presence of any defects listed in table II or any dimension not within specified requirements of the preproduction sample.

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4.3 Inspection. Sampling for inspection and acceptance shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 In-process inspection.

4.3.2.1 Process examination. Examination shall be made of the following operations to establish conformance to specified requirements. Whenever nonconformance is noted, correction shall be made to affected items and lot in process.

(a) Back splash, glass when used is heat-tempered (see 3.6.9).

(b) Thermal insulation. Installation of insulation shall be examined for compliance with 3.6.10. When nonconformance is noted correction shall be made to affected items and lot in process.

(c) Finishes. Finish operations on unexposed surfaces shall be examined for compliance with 3.9.6.

4.3.2.2 Intermediate testing. Tests shall be made to determine compliance with the requirements specified in the applicable subparagraphs of 3.9 for porcelain enamel. Test method for thickness of porcelain enamel shall be as specified in 4.4.6. The lot shall be all like porcelain enameled components; for example, cooking top fronts, doors, or aeration bowls. The sample unit shall be one component, as applicable. Acid resistance of cooking tops shall be tested as specified in ASTM C 282. The inspection levels and acceptable quality levels (AQL's) shall be as follows:

Requirement	Inspection level	AQL
Thickness of porcelain enamel	S-2	6.5
Acid resistance (cook tops only)	S-2	6.5

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4.3.2.3 Intermediate testing of knobs and handles. Knobs and handles shall be tested as specified in 4.4.5. The lot shall consist of all knobs and handles, as applicable, offered for inspection at one time. The sample unit shall be one knob or handle. The inspection level shall be S-1 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

4.3.3 Inspection of the end product. The inspection lot shall consist of all ranges of one type, size, and style offered for inspection at one time. The sample unit shall be one completely fabricated range.

4.3.3.1 Visual examination. Examination shall be made of the completely assembled range for the defects listed in table II. The inspection level shall be level II with AQL's of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Finish (general)	Not as specified	X	
	Metallic platings missing, incomplete, blistered, flaked, peeled, pitted, not uniform, or rough (when applicable)	X	
	Area of no film, runs, painted surfaces, area of rust		X
	Foreign matter imbedded in paint		X
Porcelain enamel finish (see 6.3) (for definitions of defects consult ASTM C 236)	Ground coat visible through cover coat, surface contains dark specks (when range is in the use position)	X	
	Panels of same color not uniform in depth of color or shade		X
	Chipped (exterior surface or surface not likely to contact food)		X
	Aeration bowls, burner trays, drip pans, oven racks, or any surface coming in contact with food: porcelain surface fractured, chipped, or blistered (include underside of broiler pan and top portion of oven interior liner)	X	

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TABLE II. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Porcelain enamel finish (see 6.3) (for defninitions of defects consult ASTM C 286) (con.)	Fish scaling	X	
	Copperheads accompanied by a pulling away or curling up of the enamel	X	
	Copperheads - depressed, (when range is in use position)		X
	Burn-off	X	
	Scale	X	
	Specking, crazing, hairlines, star marks, or pinholes		X
	Blisters exposing base metal	X	
	Tearing, curling, or crawling	X	
	Lump - one or more heavy lumps	X	
	Beads - any bead 1/32 inch or more thick	X	
	Scumming and lack of gloss	X	
Construction and workmanship (general)	Component part such as doors, racks, trays, drawers, broiler pans, aeration bowls, burner trays, or grates which do not fit properly	X	
	Cracked or fractured metal	X	
	Any component missing (except nuts, bolts, screws, or spring clips)	X	
	Any burr, sliver, sharp edge, rough die, tool, gouge, trim, or grind marks	X	
	Any component fractured, buckled, warped, loose, kinked, dented, punctured, or malformed	X	
	Any nut, bolt, screw, or spring clip missing		X
Baking oven and broiling section	Broiler not mounted on type of bearing specified (if drawer type)	X	
	Oven vent blocked or obstructed	X	
	Oven bottom not removable without use of tools		X

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TABLE II. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Construction and workmanship (general) (con.)			
Oven racks and broiler pan(s)	Oven racks not constructed of corrosion-resisting steel (check with magnet) porcelain coated steel, or chrome or nickel plated steel	X	
	Oven racks not provided with rack stops	X	
	Not as specified	X	
Oven and broiler door (as applicable)	Will not remain fully closed without securing	X	
	Means for adjusting tension not provided	X	
	Oven doors not recessed where required and sides of range do not project past door	X	
Cooking top	Not equipped with back splash	X	
	Back splash does not extend the full width of the cooking top	X	
Grates	Any tilt that may cause improper cooking performance, for example, range top (including grates) not horizontal with base; cooking top warped, bowed, or otherwise not level	X	
	Center of grates higher than outside periphery (check visually using straight edge)	X	

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TABLE II. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Construction and workmanship (general) (con.)			
Burners	Not at least one burner provided with thermostatic control (except for Navy procurements)	X	
	Burners not removable and replaceable	X	
	Burner head (port section) does not fit flush to body (two or three piece burner only)	X	
	A proper fit shall be determined by examining the burner assemblies against a lighted white (opaque) background with the mating surface horizontal to the eye and at a right angle to the background		
	The burner shall be rotated to provide examination of the entire periphery		
	The cap and bowl of aluminum burners shall not show any space between the mating surfaces when so examined		
	Cast iron and steel burners will be considered acceptable with a clearance not exceeding 0.010 inch between the mating surfaces of the cap and burner bowl		
Burner trays and aeration bowls (including burner bowl type)	Warped or bent aeration bowls		X
	Not removable (except when bowls are integral with top)	X	
Oven thermostat control (including tube and bulb as applicable)	Loose		X
	Marking less than 500°F.	X	
	Bake and broil positions missing or not legibly and permanently marked	X	

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TABLE II. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Construction and workmanship (general) (con.)			
Wiring, electric circuits, controls and lights	No provisions for permanently grounding the range	X	
	Oven light and switch inoperable	X	
Knobs and handles	Not plastic	X	
Control knobs	Heat positions not legible or permanent	X	
	Controls installed between burners	X	
Accessories	Not as specified	X	
Identification marking	Missing, incomplete, illegible		X

4.3.3.2 Dimensional examination. Examination shall be made of the gas ranges for compliance with dimensions specified. Any nonconformance to specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

4.3.3.3 Testing of the end item. Each range shall be tested in accordance with 4.4.1 and 4.4.7. Failure to pass the tests shall be cause for rejection of the individual range. The ranges shall be tested as specified in 4.4.2, 4.4.3 and 4.4.4 using inspection level S-3 with an AQL of 4.0 defects expressed in terms of defects per hundred units.

4.3.4 Standards compliance. Proof of compliance with the ANSI requirements of 3.3 shall be submitted to the Government representative.

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4.3.5 Examination of preparation for delivery. Examination to determine compliance with requirements for preparation for delivery shall be performed in accordance with MIL-R-11337.

4.4 Tests.

4.4.1 Operational tests. The range shall be connected to a natural or LP gas supply, as applicable. The gas shall be adjusted to within plus or minus 5 percent of the manufacturer's recommended hourly BTU input rating and the gas pressure to the burners shall be adjusted at normal pressure. The oven temperature shall be checked to determine the low temperature requirements of 3.4. The pilots and burners shall be ignited to determine conformance to 3.8. Burner valve knobs shall be turned fully open and closed to check for looseness or binding (see 3.6.4). Timers and electric clocks when provided shall be tested for proper operation to determine conformance to 3.4.2.1 and 3.4.2.2. Any nonconformance to the operational requirements specified shall constitute failure of this test.

4.4.2 Door tests. A piece of writing paper (approximately 2-1/2 inches wide) shall be held against the range body and the inside top edge of the oven door with the oven door in closed position. There shall be a definite drag on the paper when it is slowly withdrawn. This determination shall be made at various points at the top to determine compliance with the oven tightness requirements of 3.6.3. Any nonconformance to the door requirements specified shall constitute failure of this test.

4.4.3 Grate test. A container with a 20 pound load shall be placed in several positions on each grate and the vertical displacement of the grate measured to determine conformance to the grate deflection requirements of 3.6.6. Inability of the grate to meet the load requirements specified shall constitute failure of this test.

4.4.4 Oven rack test. An oven rack shall be withdrawn to the stop of any rack position and the following loads shall be distributed over the full area of the rack to determine that the rack does not deflect more than the amount specified in 3.6.2.1. Inability of the withdrawn oven rack to meet the load requirements specified shall constitute failure of this test.

Width of rack (inches)	Load (pounds)
Up to 14	15
14 to 18	20
18 and over	25

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4.4.5 Plastic tests.

4.4.5.1 Self-extinguishing test. The plastic knobs shall be tested as specified in ASTM Method D 635 to determine conformance to flammability requirements of 3.6.8. Inability to meet the requirements specified shall constitute failure of this test.

4.4.5.2 Drop test. The plastic gas cock knobs and control valve knobs on the sample range shall be dropped three times from a height of 4 feet onto an asphalt or vinyl tile surface over a cement floor to determine compliance with strength requirements of 3.6.8. Any nonconformance to the requirements specified shall constitute failure of this test.

4.4.6 Porcelain enamel thickness. Porcelain enamel thickness shall be measured with a magnetic thickness gauge calibrated to an accuracy of plus or minus 0.001 inch to determine conformance to finish thickness requirements of applicable subparagraphs of 3.9. Any nonconformance to the requirements specified shall constitute failure of this test.

4.4.7 Gas leakage. The range shall be tested for leaks in the fuel system while the range is connected to a gas supply line to determine conformance to leakage requirements of 3.6.4. Inability to meet the leakage requirements specified shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging and packing for level A, B, or C shall be in accordance with MIL-R-11337 (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-R-11337.

6. NOTES

6.1 Intended use. The ranges are intended for use in domestic kitchens of civil and military quarters.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification
(b) Type, size, and style range required (see 1.2).
(c) First article (see 3.1, 4.2 and 6.4).
(d) Special features required for type I, styles 1 and 2 ranges (see 3.4.2.1).
(e) When procurement is for Navy (see 3.6.5).
(f) When procurement is for Far East or Pacific Ocean locations (see 3.6.5).
(g) Electrical requirements, if other than specified (see 3.6.12).
(h) Selection of applicable levels of preservation, packaging, and packing (see 5.1).

6.3 Definitions. For definitions of porcelain enamel defects consult ASTM Publication C 286 - Terms Relating to Porcelain Enamel.

6.4 First article. The items covered by this specification required first article test and approval under the appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulations. The first article should be a preproduction sample gas range. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examination, test, and approval of the first article.

6.5 Manuals. Any requirments for technical manuals for the item covered by this specification should be included in DD Form 1423 Contract Data Requirements List and cited in the contract.

6.6 Supersession data. Items identified herein were class 1, design 2, and style 2 listed under "Types I and II" of Federal Specification S-R-44c. Items identified in S-R-44c as type I, size 40 and class 2, design 1, style 1, and models 1 and 2 were deleted. Two new styles have been established in this revision as indicated in 1.2. This specification includes the requirements of the Department of the Navy, Naval Facilities Engineering Command Purchase Description, CBC PD 7310-1, dated 20 May 1969.

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Navy - YD
Air Force - 84

Review activities:

Army - MD, CE
Navy - MC

Preparing activity:

Army - GL

CIVIL AGENCY INTEREST:

GSA-FSS
HEW

Project No. 7310-0279

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

S-R-44D Ranges, Gas (Domestic)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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